

A Paradoxical, Self-Referential System of Logics

Four non-classical logics as resolutions of one universal diagonal,
with two machine-checked poles

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Lean 4 witnesses: github.com/inventor1975/ZTL (`lean/ZTL.lean`, `lean/QuantumWitness.lean`)

Verify from zero: §6 — two files, two commands, no dependencies

Abstract

We propose a reading of four non-classical logics — classical, intuitionistic / type-theoretic, quantum (orthomodular), and Zero-Trust Logic (ZTL) — as four disciplined responses to a single self-referential construction: the universal diagonal underlying the liar, Russell, Cantor, Gödel and Tarski (Lawvere 1969; Yanofsky 2003). Each retains all classical structural principles *save one*, and each thereby houses the diagonal differently. Two features are invariant across all four: non-contradiction with its associated reductio (a retained *principle*), and the diagonal itself (a retained *obstruction*). We isolate the one part of this picture that is formal rather than expository — the complementary sacrifice made by ZTL and by orthomodular quantum logic — and record it as two propositions machine-checked in Lean 4 on the *empty axiom list* (no classical choice, no quotient, no propositional extensionality). We are explicit throughout about what is established (all four logics; the Lawvere–Yanofsky unification), what is the author’s (ZTL; the arrangement; the two witnesses), and what is analogy carrying no proof (the “cycle” itself). This is a synthesis and a reading, not a theorem, a merger, a new foundation, or a new field.

Keywords. self-reference; Lawvere fixed point; diagonal argument; quantum logic; orthomodular lattice; three-valued logic; paracomplete logic; Zero-Trust Logic; universal logic; Lean 4; machine-checked proof.

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1 Scope

The claim of this note is deliberately narrow. It is *not* that a new logic or a new foundation has been discovered, nor that the four logics below are amalgamated into one calculus — they are provably incompatible in the precise sense of §5. The unification of the classical antinomies under a single fixed-point schema is due to Lawvere [7] and, in the concrete set-theoretic form used here, to Yanofsky [11]; quantum logic is Birkhoff–von Neumann [1]; the intuitionistic corner is standard; the “general theory of logical systems” already exists as a field (universal logic, abstract model theory [13]). The single component original to the author is **ZTL** [12], which appears here only as one corner. What this note adds is (i) the *arrangement* of the four as four resolutions of one diagonal, and (ii) two machine-checked propositions (§6) exhibiting the sharpest pair of them as complementary. Everything expository is marked as such; everything proved is in §6 and reproducible from zero.

2 The universal diagonal

The classical antinomies are one construction in several presentations. Lawvere’s fixed-point theorem makes this exact.

Lemma 2.1 (Lawvere [7]). *In a cartesian closed category, if there is a point-surjective morphism $\varphi : A \rightarrow Y^A$, then every endomorphism $g : Y \rightarrow Y$ has a fixed point $y : 1 \rightarrow Y$ with $g \circ y = y$.*

Contrapositively: a *fixed-point-free* endomorphism of Y obstructs any point-surjection $A \rightarrow Y^A$. Instantiating Y and g :

- $Y = 2$, $g = \neg$ (no fixed point): no surjection $A \rightarrow 2^A$ — **Cantor**;
- the same schema, read in an internal truth object with a fixed-point-free negation, yields the **liar** and **Russell**;
- read in a provability / representation setting it yields **Gödel** and **Tarski**’s undefinability (Yanofsky [11], §§3–5).

The common core is *self-application against a fixed-point-free map*: an object that names its own maps, composed with a g that has no fixed point. Following Yanofsky, we call this schema **the diagonal**. It is *pre-classical*: it is not built by classical logic but is a construction that any logic expressive enough to talk about its own sentences must confront. A logic that ignores it does not escape it; it trivializes on it.

3 Four resolutions

A logic admitting self-reference must do something disciplined with the diagonal or collapse. There are (at least) four such disciplines in current use, and each is *classical logic minus exactly one structural principle*:

Logic	Principle dropped	Resolution of the diagonal
Classical	none	explodes on the raw liar, then localizes truth one level up (Tarski hierarchy)
Intuitionistic / type-theoretic	excluded middle; unrestricted Type : Type	bans self-typing — the universe hierarchy; Girard’s paradox is the diagonal of naïve type theory (Coquand [3]; Hurkens [4])
Quantum (orthomodular)	distributivity	the self-negating join lives as a genuine element; complementation stays total
ZTL	identity $p \rightarrow p$, excluded middle, double negation	pointwise quarantine: the liar has no fixed value; it is flagged, not exploded [12]

Two honesty points. First, only the ZTL row is the author’s; the other three are the standard content of their fields, curated here, not discovered. Second, the classical row is the informative one: classical logic retains *every* structural principle, which is exactly why it is the system that must *explode* on the unlocalized liar and repair itself from the metalanguage. Retaining everything is paid for in the hierarchy.

4 Two invariants

Across the four, two things do not move — a dual pair, one positive and one negative.

The floor (a retained principle). Non-contradiction and the reductio it licenses, $(p \rightarrow \neg p) \rightarrow \neg p$, survive in every corner. In the quantum lattice $x \wedge x^\perp = 0$; in the ZTL matrix $\neg(p \wedge \neg p)$ is designated for every value; both are among the propositions of §6.

The ceiling (a retained obstruction). The diagonal survives in every corner, precisely because each logic is expressive enough to state it. What differs is only the *manner of housing* it (§3).

The two are linked: **non-contradiction is the guard against the diagonal**. A logic that surrendered non-contradiction would not tame the diagonal but be trivialized by it (*ex contradictione quodlibet*). Where the obstruction persists, the floor stands; that is why both are invariant.

5 No consistent amalgamation

Why an irreducible plurality rather than a hierarchy or a common refinement?

Because the retained principles are *pairwise incompatible*. Quantum logic retains $x \vee x^\perp = 1$; ZTL withholds $p \vee \neg p$ at the mark — no single valuation structure validates both a total complementation law and its pointwise failure. Type theory demands strong normalization (every term terminates); the ZTL solver houses non-termination as the transient phase **N** (§6). Any system retaining *all four* families of principles at once would validate distributivity *and* its negation, hence be trivial — that system is classical logic, which pays for retaining everything with the raw diagonal it must then hierarchically contain. There is therefore no consistent common refinement; the four are genuinely incomparable.

We describe this informally as an *orbit* around classical logic — the unreachable centre that keeps all principles at the price of the liar. We stress that “orbit”, “centre” and “cycle” are metaphor: the content is the pairwise incompatibility above and the concrete complementarity of §6. No claim of a formal cyclic structure is made.

6 The ZTL–quantum mirror, machine-checked

The one formalizable part of the picture is the *complementary sacrifice* made by ZTL and by orthomodular logic. Both are recorded below as propositions verified in Lean 4 over finite carriers by `decide`, on the **empty axiom list**, in bare Lean (no `mathlib`, no imports).

Pole A (ZTL). Let $M = \{\mathsf{T}, \mathsf{F}, \mathsf{Z}\}$, with connectives generated by the *zero-trust lift*: for a Boolean f , $\text{lift}(f)(\vec{x}) = \mathsf{T}$ iff f returns `true` under every classical reading of the arguments (Z read as both `true` and `false`), else F . Then in M :

Proposition 6.1 (`lean/ZTL.lean`, empty axiom list).

- (i) non-contradiction — $\neg(p \wedge \neg p) = \mathsf{T}$ for all p ;
- (ii) distributivity in both directions — $p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$ and dually, for all p, q, r ;
- (iii) failure of identity, excluded middle and double negation — $p \rightarrow p$, $p \vee \neg p$, $\neg\neg p \rightarrow p$ are not designated for all p (each is falsified at $p = \mathsf{Z}$).

Here the third symbol **Z** is a **status mark, not a truth value**: verdicts are two-valued, **Z** marking an unverified input (greedy register) or, reused positionally in the solver, the transient phase **N** = “not yet” (Kleene’s undefined), which never labels an input. ZTL is thus **paracomplete** (truth-value gaps, no gluts), not paraconsistent [12].

Pole B (orthomodular). Let MO_2 be the smallest non-distributive orthomodular lattice: $\{0, 1\}$ together with two complementary pairs $(a, a^\perp), (b, b^\perp)$, all four proper elements mutually incompatible, with $x \wedge x^\perp = 0, x \vee x^\perp = 1$. Then:

Proposition 6.2 (`lean/QuantumWitness.lean`, empty axiom list).

- (i) non-contradiction — $x \wedge x^\perp = 0$ for all x ;
- (ii) excluded middle and double negation — $x \vee x^\perp = 1$ and $x^{\perp\perp} = x$ for all x ;
- (iii) failure of distributivity — $a \wedge (b \vee b^\perp) = a \wedge 1 = a$, whereas $(a \wedge b) \vee (a \wedge b^\perp) = 0 \vee 0 = 0$.

The mirror. ZTL retains distributivity and surrenders excluded middle, double negation and identity; orthomodular logic retains excluded middle and double negation and surrenders distributivity; both retain non-contradiction. Informally: ZTL fails where a proposition meets *itself*; quantum logic fails where propositions *combine*.

Remark (the honest mark). The two “retentions of excluded middle” are not instances of one theorem in one system. In MO_2 , $x \vee x^\perp = 1$ is a *lattice identity* and $x^{\perp\perp} = x$ holds because the orthocomplement is involutive *by the choice of an ortholattice* — mild, structural. In M , the *failure* of $p \vee \neg p$ is a computation in a *logical matrix*, where validity means designation. These are facts in two different formalisms that rhyme; the “mirror” is an analogy made precise on each side, not a single cross-formalism theorem. This is the boundary between what is proved (each proposition) and what is read (their pairing).

Reproduction (Lean 4.29.1, measured 2026-07-18):

```
$ lean lean/ZTL.lean          # 13 objects, no axioms,  ~1.1 s
$ lean lean/QuantumWitness.lean #  5 objects, no axioms,  ~0.5 s
```

Every `#print axioms` line reports “*does not depend on any axioms*”, definitions included: no `Classical.choice`, no `Quot.sound`, no `propext`. The intuitionistic and classical corners are, by contrast, *metatheorems* (underivability of excluded middle; failure of strong normalization for `Type : Type`; “classical as centre”) and are not formalized here by design; the diagonal itself, ranging over four logics that are not objects of one theory, is likewise not a single formal object.

7 Provenance

	Content	Status
Established	Lawvere’s fixed-point theorem and the diagonal unification (Lawvere [7]; Yanofsky [11]); quantum logic and non-distributivity (Birkhoff–von Neumann [1]); intuitionism, the universe hierarchy, Girard’s paradox (Martin-Löf [9]; Coquand [3]; Hurkens [4]); universal logic as a field (Béziau [13])	literature
Author’s	ZTL, the zero-trust paracomplete logic [12]; the arrangement of the four as resolutions of one diagonal; the two Lean witnesses on the empty axiom list	synthesis + machine-checked
Analogy (no proof)	the “cycle / orbit / centre”; “the floor guards the ceiling”; “each logic pays one principle so none pays the whole diagonal”	structural metaphor, marked

The defensible claim is narrow: four established non-classical logics align as four disciplined resolutions of one pre-classical diagonal, and the sharpest pair among them — ZTL and orthomodular logic — make complementary, machine-checkable sacrifices. The organizing picture is offered as a reading; its two poles are offered as proof.

References

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AI disclosure. This note was written in dialogue with the AI system Claude (Anthropic); all design decisions, the framing and final responsibility are the human author’s. The reliability of §6 depends on neither the author nor the AI: both Lean files verify against the Lean 4 kernel on the empty axiom list. Everything outside §6 is offered as a reading and marked as such.

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